GATE ME 90 Days Challenge

Day 20/90 · Practice Set with Detailed Solutions

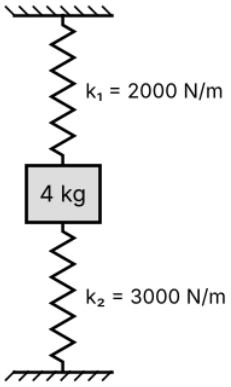
SUBJECT	TODAY'S TOPIC	COUNTDOWN
Vibrations	Free Vibration	88 days to GATE

How to use this set: 15 questions (5 MCQ, 10 NAT) on undamped free vibration of single-DOF systems (equivalent springs, energy and Rayleigh methods, torsional and pendulum systems), viscous damping, damping ratio, logarithmic decrement and Coulomb damping. Try them in **40 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 NAT

A 4 kg block is held between two vertical springs, k_1 = 2000 N/m above it and k_2 = 3000 N/m below it, both ends fixed, as shown. The natural frequency of vertical vibration (in Hz, round off to two decimal places) is _____



Q2 MCQ

The static deflection of a machine on its spring mounts is increased four times (by using softer springs). Its natural frequency becomes

(A) four times
(B) twice
(C) one-half
(D) one-quarter

Q3 NAT

A helical spring of stiffness 3000 N/m has a mass of 1.5 kg and carries a 6 kg mass at its lower end. Accounting for the inertia of the spring (Rayleigh's method), the natural frequency (in rad/s, round off to two decimal places) is _____

Q4 NAT

A U-tube manometer contains a liquid column of total length 0.6 m. If the liquid is disturbed, it oscillates in the tube. Neglecting friction, the natural circular frequency of oscillation (in rad/s, round off to two decimal places) is _____

Q5 MCQ

A cylinder floats upright in water with an immersed depth d. When pushed down slightly and released, it bobs up and down. Neglecting the added mass of water and damping, its natural frequency is

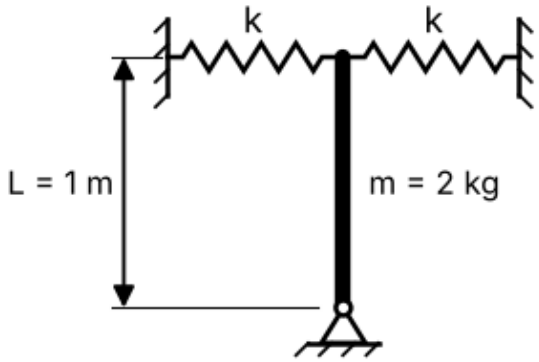
(A) $\sqrt{(g/d)}$
(B) $\sqrt{(2g/d)}$
(C) $\sqrt{(d/g)}$
(D) $\sqrt{(g/2d)}$

Q6 NAT

A disc with mass moment of inertia 0.02 kg·m² hangs from a steel wire of diameter 2 mm and length 0.8 m (G = 80 GPa). The period of torsional oscillation (in s, round off to two decimal places) is _____

Q7 NAT

A uniform bar of mass 2 kg and length 1 m is hinged at its lower end and stands vertical. Two horizontal springs, each of stiffness 500 N/m, are attached at its top end, as shown. The natural frequency of small oscillations (in rad/s, round off to two decimal places) is _____



Q8 MCQ

A spring-mass-damper system has m = 2 kg, k = 800 N/m and c = 40 N·s/m. The system is

(A) critically damped
(B) overdamped with $\zeta = 2$
(C) underdamped with $\zeta = 0.5$
(D) underdamped with $\zeta = 0.05$

Q9 NAT

In a free vibration test the amplitude falls from 12 mm to 3 mm in 4 complete cycles. The damping ratio (round off to four decimal places) is _____

Q10 NAT

A system with m = 8 kg and k = 12.8 kN/m is to have a damping ratio of 0.25. The required damping coefficient (in N·s/m) is _____

Q11 MCQ

Door closers and the recoil mechanisms of guns are designed to be close to critically damped because such a system

(A) oscillates many times with a small amplitude

(B) returns to its equilibrium position in the shortest time without overshooting

(C) never returns to equilibrium

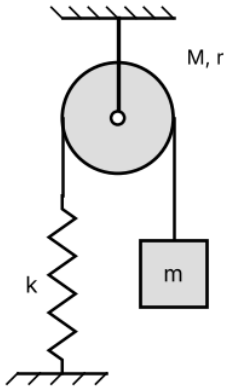
(D) has the highest natural frequency

Q12 NAT

A 5 kg block on a rough horizontal surface ($\mu = 0.15$) is attached to a spring of stiffness 2000 N/m. It is pulled 50 mm from the unstretched position and released. Taking $g = 9.81$ m/s², the amplitude after 2 complete cycles (in mm, round off to two decimal places) is _____

Q13 NAT

A 2 kg mass hangs from a light inextensible rope passing over a pulley (a solid disc of mass 4 kg and radius 100 mm, frictionless bearing). The other end of the rope is tied to a spring of stiffness 1200 N/m anchored to the ground, as shown. If the rope does not slip on the pulley, the natural frequency (in rad/s, round off to two decimal places) is _____



Q14 MCQ

A light simply supported beam of span L and flexural rigidity EI carries a mass m at mid-span. The natural frequency of transverse vibration is

(A) $\sqrt{(3EI/(mL^3))}$

(B) $\sqrt{(192EI/(mL^3))}$

(C) $\sqrt{(384EI/(5mL^3))}$

(D) $\sqrt{(48EI/(mL^3))}$

Q15 NAT

A lightly damped structure has damping ratio 0.02. The number of cycles needed for the free-vibration amplitude to fall to half its initial value (round off to two decimal places) is _____

Answer Key

Q	Type	Answer
Q1	NAT	5.63 (accept 5.59 to 5.66)
Q2	MCQ	(C) one-half
Q3	NAT	21.48 (accept 21.35 to 21.61)
Q4	NAT	5.72 (accept 5.68 to 5.75)
Q5	MCQ	(A) $\sqrt{(g/d)}$
Q6	NAT	2.24 (accept 2.23 to 2.26)
Q7	NAT	38.54 (accept 38.31 to 38.77)
Q8	MCQ	(C) underdamped with $\zeta = 0.5$
Q9	NAT	0.0551 (accept 0.0545 to 0.0556)
Q10	NAT	160
Q11	MCQ	(B) returns to its equilibrium position in the shortest time without overshooting
Q12	NAT	20.57 (accept 20.45 to 20.69)
Q13	NAT	17.32 (accept 17.22 to 17.42)
Q14	MCQ	(D) $\sqrt{(48EI/(mL^3))}$
Q15	NAT	5.51 (accept 5.46 to 5.57)

Detailed Solutions

Q1 **Answer: 5.63 (accept 5.59 to 5.66)**

Concept: When the block moves down by x, the upper spring stretches by x and the lower spring compresses by x. Both forces oppose the motion, so the springs act **in parallel** even though they are on opposite sides.

$k_e = k_1 + k_2 =$ **5000 N/m**

$\omega_n = \sqrt{(5000/4)} =$ **35.36 rad/s** $\rightarrow f_n = \omega_n/2\pi =$ **5.63 Hz**

Exam tip: Ask "do the springs share the same displacement or the same force?" Same displacement → parallel (add k). Same force → series (add 1/k).

Q2 **Answer: (C) one-half**

Concept: $\omega_n = \sqrt{(k/m)} = \sqrt{(g/\delta_{st})}$, since $mg = k\delta_{st}$.

So $f_n \propto 1/\sqrt{\delta_{st}}$; $\delta_{st} \times 4 \rightarrow f_n \times 1/2$.

Exam tip: Handy shortcut: f_n (Hz) $\approx 15.76/\sqrt{\delta_{st}}$ with δ_{st} in mm. For 1 mm deflection, $f_n \approx 15.8$ Hz.

Q3 **Answer: 21.48 (accept 21.35 to 21.61)**

Concept: Assuming a linear velocity profile along the spring, its kinetic energy equals that of one-third of its mass moving with the end mass.

$m_{eff} = m + m_s/3 = 6 + 0.5 =$ **6.5 kg**

$\omega_n = \sqrt{(3000/6.5)} =$ **21.48 rad/s** (ignoring the spring mass would give 22.36 rad/s)

Exam tip: Spring mass adds $m_s/3$; for a cantilever beam the effective end mass is 33/140 of the beam mass, and for a simply supported beam (central mass) it is 17/35.

Q4 **Answer: 5.72 (accept 5.68 to 5.75)**

Concept: Displace the liquid by x on one side. The level difference is 2x, so the restoring force is ρgA(2x), acting on the whole column of mass ρAL.

$\rho AL\,\ddot{x} + 2\rho gA\,x = 0 \rightarrow \omega_n = \sqrt{(2g/L)}$

$\omega_n = \sqrt{(2 \times 9.81/0.6)} =$ **5.72 rad/s**

Exam tip: The result is independent of density and tube area. It equals a simple pendulum of length L/2.

Q5 **Answer: (A) $\sqrt{(g/d)}$**

Concept: An extra depth x adds buoyancy $\rho_w gAx$, so the stiffness is $k = \rho_w gA$.

In equilibrium the weight is $m\,g = \rho_w gAd$, so $m = \rho_w Ad$.

$\omega_n = \sqrt{(k/m)} = \sqrt{(\rho_w gA/(\rho_w Ad))} =$ **$\sqrt{(g/d)}$**

Exam tip: It is the same $\sqrt{(g/\delta)}$ rule again: the immersed depth plays the role of the static deflection.

Q6 **Answer: 2.24 (accept 2.23 to 2.26)**

Concept: Torsional stiffness of a shaft or wire: $k_t = GJ/L$, with $J = \pi d^4/32$.

$J = \pi(0.002)^4/32 = 1.5708 \times 10^{-12}\ m^4 \rightarrow k_t = 80\times10^9 \times 1.5708\times10^{-12}/0.8 =$ **0.1571 N·m/rad**

$\omega_n = \sqrt{(k_t/I)} =$ **2.802 rad/s** $\rightarrow T = 2\pi/\omega_n =$ **2.24 s**

Exam tip: This is how moments of inertia are measured in the lab: time the torsional pendulum, then $I = k_t T^2/(4\pi^2)$.

Q7 **Answer: 38.54 (accept 38.31 to 38.77)**

Concept: For an inverted pendulum gravity is **destabilising**: it adds a negative stiffness $-mg(L/2)$ to the springs' restoring moment.

Restoring moment for a small tilt θ: $(2kL^2 - mgL/2)\theta = (1000 - 9.81)\theta =$ **990.19 θ** N·m

$I_O = mL^2/3 = 0.6667$ kg·m² $\rightarrow \omega_n = \sqrt{(990.19/0.6667)} =$ **38.54 rad/s**

Exam tip: The system is stable only if $2kL^2 > mgL/2$. Hanging (pendulum below the hinge), the gravity term would add instead of subtract.

Q8 **Answer: (C) underdamped with ζ = 0.5**

Concept: $\zeta = c/c_c$ with $c_c = 2\sqrt{(km)} = 2m\omega_n$.

$c_c = 2\sqrt{(800 \times 2)} = \mathbf{80\ N{\cdot}s/m} \rightarrow \zeta = 40/80 = \mathbf{0.5}$ (< 1, underdamped)

Exam tip: $\omega_n = 20$ rad/s and $\omega_d = 20\sqrt{(1 - 0.25)} = 17.32$ rad/s: the system still oscillates, but the amplitude dies quickly.

Q9 **Answer: 0.0551 (accept 0.0545 to 0.0556)**

Concept: Logarithmic decrement over n cycles: $\delta = (1/n)\ln(x_0/x_n)$; then $\zeta = \delta/\sqrt{(4\pi^2 + \delta^2)}$.

$\delta = (1/4)\ln(12/3) = (1/4)\ln 4 = \mathbf{0.3466}$

$\zeta = 0.3466/\sqrt{(39.478 + 0.1201)} = \mathbf{0.0551}$

Exam tip: For small damping $\zeta \approx \delta/2\pi = 0.0552$: the approximation is good to better than 0.2 % here.

Q10 **Answer: 160**

Concept: $c = \zeta\, c_c = \zeta \times 2\sqrt{(km)}$.

$c_c = 2\sqrt{(12800 \times 8)} = 2 \times 320 = \mathbf{640\ N{\cdot}s/m}$

$c = 0.25 \times 640 = \mathbf{160\ N{\cdot}s/m}$

Exam tip: Write $c_c = 2m\omega_n$: here $\omega_n = 40$ rad/s, so $c_c = 2 \times 8 \times 40 = 640$ N·s/m. Same number, often faster.

Q11 **Answer: (B) returns to its equilibrium position in the shortest time without overshooting**

Concept: Underdamped systems overshoot and oscillate; overdamped systems creep back slowly. Critical damping ($\zeta = 1$) is the fastest non-oscillatory return.

$x(t) = (A + Bt)e^{-\omega_n t}$ for $\zeta = 1$: no oscillatory term.

Exam tip: Measuring instruments (galvanometers, pointer gauges) use $\zeta \approx 0.6$ to 0.7 instead: a small overshoot buys a faster settling time.

Q12 **Answer: 20.57 (accept 20.45 to 20.69)**

Concept: Coulomb damping: the friction force $F = \mu mg$ is constant, so the amplitude drops by $2F/k$ every half-cycle, i.e. $4F/k$ per cycle (a straight-line decay).

$F = 0.15 \times 5 \times 9.81 = \mathbf{7.3575\ N}$; $4F/k = \mathbf{14.715\ mm\ per\ cycle}$

After 2 cycles: $50 - 2 \times 14.715 = \mathbf{20.57\ mm}$ (still larger than $F/k = 3.68$ mm, so it keeps moving)

Exam tip: Viscous damping: amplitudes fall in a geometric ratio. Coulomb damping: they fall by equal steps. The shape of the decay curve tells you the damping type.

Q13 **Answer: 17.32 (accept 17.22 to 17.42)**

Concept: Energy method. Rope does not slip, so the pulley turns at $\dot{\theta} = \dot{x}/r$ and the spring stretches by x.

$KE = \tfrac{1}{2}m\dot{x}^2 + \tfrac{1}{2}(\tfrac{1}{2}Mr^2)(\dot{x}/r)^2 = \tfrac{1}{2}(m + M/2)\dot{x}^2$; $PE = \tfrac{1}{2}kx^2$

$m_{eff} = 2 + 4/2 = \mathbf{4\ kg} \rightarrow \omega_n = \sqrt{(1200/4)} = \mathbf{17.32\ rad/s}$

Exam tip: A solid-disc pulley adds half its mass to the moving mass; a ring-type pulley ($I = Mr^2$) would add its full mass.

Q14 **Answer: (D) $\sqrt{(48EI/(mL^3))}$**

Concept: The beam acts as a spring of stiffness $k = P/\delta$ for a load at the mass position.

Central load on a simply supported beam: $\delta = PL^3/(48EI) \rightarrow k = \mathbf{48EI/L^3}$

$\omega_n = \sqrt{(k/m)} = \mathbf{\sqrt{(48EI/(mL^3))}}$

Exam tip: Remember the four: cantilever end load $3EI/L^3$; simply supported centre $48EI/L^3$; fixed-fixed centre $192EI/L^3$; propped cantilever centre $768EI/(7L^3)$.

Q15 **Answer: 5.51 (accept 5.46 to 5.57)**

Concept: After n cycles $x_n = x_0 e^{-n\delta}$, with $\delta = 2\pi\zeta/\sqrt{(1 - \zeta^2)}$.

$\delta = 2\pi(0.02)/\sqrt{(1 - 0.0004)} = \mathbf{0.12569}$

$n = \ln 2/\delta = 0.69315/0.12569 = \mathbf{5.51\ cycles}$

Exam tip: For small ζ, $n_{1/2} \approx 0.11/\zeta$. Half-amplitude in about 5.5 cycles means $\zeta \approx 0.02$, a quick estimate from a recorded trace.

What next?

Day 20 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

Website test series

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Next, Day 21/90: General Aptitude, Quantitative Aptitude I. Comment **GATE90** on our reel to get all links. Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.